Aditya Krishnan

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SUMMARY

ML engineer and published researcher (NeurIPS, STOC, ICML) with expertise in building high-performance indexing and data processing infrastructure in Rust and Python for ML workloads as well as training retrieval models for RAG workflows.

PROFESSIONAL EXPERIENCE

Microsoft — M365 Copilot, Semantic Search — Remote | Mar 2025 – Present

Senior Software Engineer

- Designed and deployed a data encoding algorithm for embeddings that reduced DRAM and SSD footprint by over 2× across Substrate's serving fleet, delivering $X millions in annual infrastructure cost savings for M365 Copilot's semantic search.
- Leading research evaluation of data representation strategies — multi-vector encodings and passage embedding models — measuring their impact on downstream semantic quality and compute efficiency.
- Core contributor to DiskANN — Microsoft's open-source graph-based ANN search library built in Rust. Implementing the graph search algorithm against SSD-based persistence and optimizing index serve using SIMD kernels.

Pinecone Systems — New York City | Jan 2024 – Oct 2024

Senior Research Scientist

- Trained a production-grade neural reranker with a DeBERTa backbone using a novel negative sampling strategy. Managed the full ML pipeline — data curation, distributed training (PyTorch DDP), hyperparameter search, and evaluation — outperforming Cohere-Rerank-V3 on passage retrieval benchmarks.
- Developed a novel query-routing mechanism for IVF-style vector indexes, reducing read overhead of Pinecone's query serve engine by 25% (work published in NeurIPS 2025).

Pinecone Systems — New York City | Oct 2022 – Dec 2023

Research Scientist

- Built Pinecone's 1st-place solution for the NeurIPS 2023 Big ANN Benchmark (OOD track) in Rust — achieving 35K+ QPS on 10M vectors with 8 vCPUs and 16 GB RAM through novel data layout and optimized SIMD kernels.

EDUCATION

Johns Hopkins University, Whiting School of Engineering — Baltimore, MD | Sep 2018 – Sep 2022

Ph.D. in Computer Science — Advisor: Vladimir Braverman

Thesis: Fast and Memory-Efficient Algorithms for Matrix Spectrum Approximation

Carnegie Mellon University, School of Computer Science — Pittsburgh, PA | Aug 2013 – May 2018

B.S. and M.S. in Computer Science

TECHNICAL SKILLS

Languages and Frameworks: Rust, Python, PyTorch, Distributed Data Parallel, FSDP

Infrastructure: Large-scale ML data pipelines and algorithms, encoding systems, performance benchmarking, SIMD intrinsics

SELECTED PUBLICATIONS

Authors appear in alphabetical order. Full list on Google Scholar.

NeurIPS 2025: *Optimistic Query Routing for Maximum Inner-Product Search*. With S. Bruch and F.M. Nardini

STOC 2022: *Sublinear Time Spectral Density Estimation*. With V. Braverman and C. Musco

ACML 2021: *Lifelong Learning with Sketched Structural Regularization*. With H. Li, J. Wu, S. Kolouri, P.K. Pilly, and V. Braverman

ICML 2020: *Schatten Norms in Matrix Streams: Hello Sparsity, Goodbye Dimension*. With V. Braverman, R. Krauthgamer, and R. Sinoff

HONORS & AWARDS

- NeurIPS Top Reviewer - awarded to < 10% of reviewers (2022)
- JHU MINDS TRIPODS Data Science Fellowship (2022)
- JHU CS Department Fellowship — awarded to 2 of 50+ incoming PhD students (2018)